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Some Recent Palaeomagnetic Measurements made at Imperial College, London

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Introduction

This paper gives a brief report of some recent palaeomagnetic measurements made by the Group working in the Physics Department at the Imperial College of Science and Technology, London. The present research programme includes investigations of Triassic red sandstones from France and Spain, of lavas from the Deccan Traps in India, and of igneous rocks and baked sediments, mostly Carboniferous in age, from a number of areas in Britain.

§ 1. British Igneous Rocks and Baked Sediments

The study of the baked sediments found in close contact with igneous rocks is one of considerable palaeomagnetic interest, and a large number of baked sediments and associated igneous rocks have been obtained from Scotland and, in collaboration with Mr. J. C. Belshé of the Department of Geodesy and Geophysics Cambridge, from various sites in the English Midlands. The collection includes :

(*a*) Baked and igneous rocks from six sites extending over a distance of 30 miles between the Clee Hills and Walsall, probably belonging to the late Carboniferous,

(*b*) Baked rocks associated with Upper Carboniferous intrusions from Tideswelldale in Derbyshire, and

(*c*) A series of lava flows, Lower Carboniferous in age, exposed along the shore at Kinghorn, Fifeshire. These latter beds are numbered consecutively 1–65 from the bottom upwards in the following text.

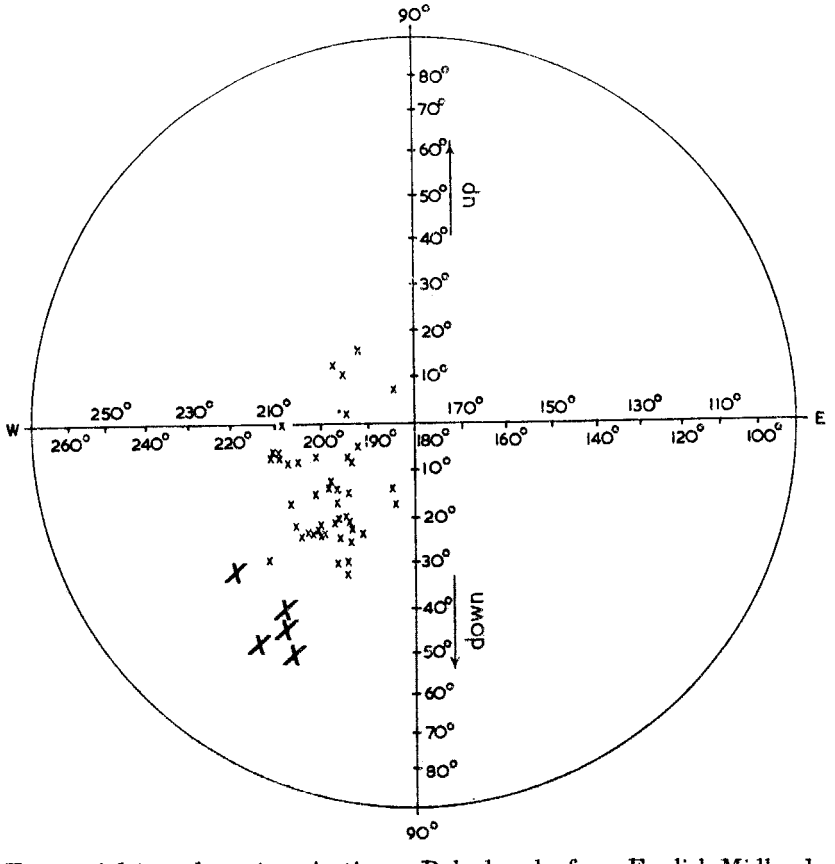
Table 1

	Declination	Dip
Clee Hill	N 200° E	+15° down
Tideswelldale	N 218° E	+36° down
Kinghorn Flows 41–46	N 26° E	−42° up
,, 48–54	N 200° E	+38° down
,, 64–65	N 20° E	+15° down

The results are plotted in figs. 1 and 2 and the mean directions of magnetization are shown in table 1. It is evident that the rocks from the English Midlands (Clee Hill and Tideswelldale) have a marked preferential

axis of magnetization, dipping downwards in a direction some 20° to the East of South. The Scottish lavas, with the exception of the two uppermost beds (64 and 65) are magnetized along a similar axis, but have steeper dips. There is a magnetic reversal in the middle of the succession. The fact that beds 64 and 65 are magnetized in a direction much closer to the present geomagnetic field than the remainder of the rocks sampled may be due to instability, and the results as a whole can evidently best be explained on the supposition that Britain, during the Carboniferous Period, occupied a position to the South of the equator.

Fig. 1

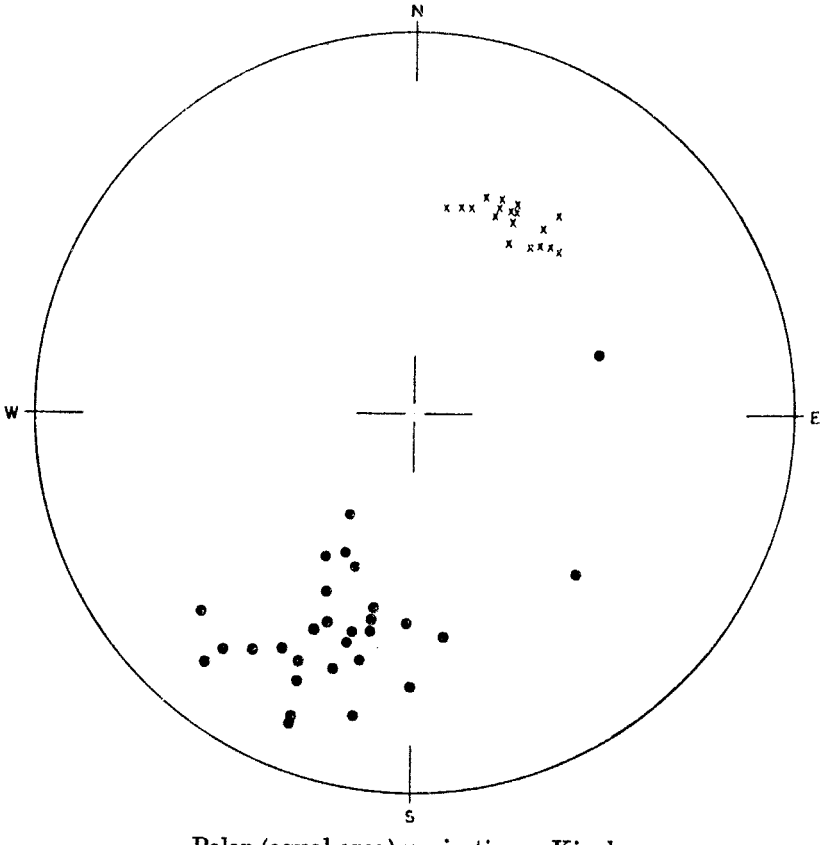


Equatorial (equal area) projection. Baked rocks from English Midlands.
× From Clee Hill and Walsall area.
× From Tideswelldale.

It is disturbing to compare these measurements with those made by J. C. Belshé (1957), in an extensive survey of British Carboniferous rocks reported elsewhere in this volume. Belshé examined a large number of unbaked sediments and a few igneous rocks, and found the mean direction of magnetization to be N 26° E with a downward dip

of 37°. This differs by rather more than 50° from the directions given in table 1 for the English rocks, and by about 80° from those given for the Scottish lavas. Such a discrepancy is hard to explain. Consideration of the time scales involved evidently precludes the possibility of its being due to large secular variation, for Belshé's collection and ours each included rocks differing in age by millions of years. It could arise from the incorrect dating of some of the formations sampled, but this, in view of the reliable geological evidence, seems to be very unlikely.

Fig. 2



Polar (equal area) projection. Kinghorn.
× Flows 41–46 (upwards).
● Flows 50–54 (downwards).

We appear to be left, then, with four possible explanations—viz :

(*a*) That some of the rocks acquired their present magnetization at a time later than the Carboniferous. It is hardly conceivable, however, that this can be true of the baked sediments, for the fact that the intensity of magnetization invariably falls off with distance from the contact provides overwhelming evidence to suggest that the magnetism was acquired at the time of emplacement of the molten magma.

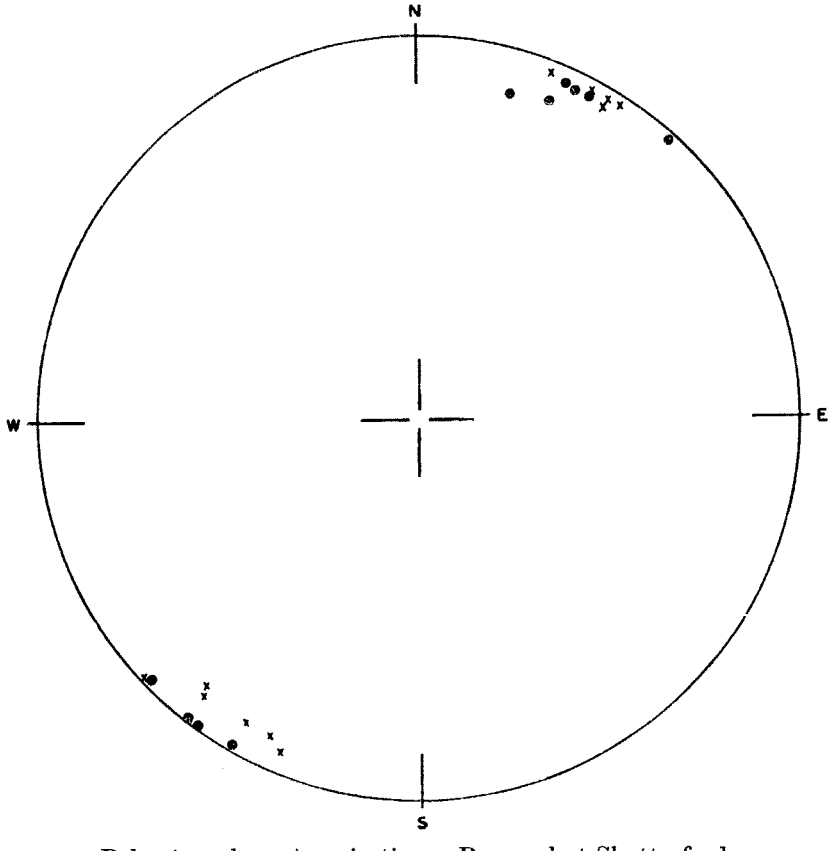
(*b*) That by some unknown physiochemical process some of the rocks acquired, at the time when they were magnetized, magnetic polarizations in a direction oblique to that of the Earth's field.

(*c*) That a large change occurred in the position of the British land mass relative to the poles at some time during the Carboniferous Period.

(*d*) That over a period of millions of years the Earth's field had a persistent mean direction which differed substantially from that of an axial dipole.

Further work on igneous and baked rocks of the Carboniferous Period from other British sites is in progress.

Fig. 3

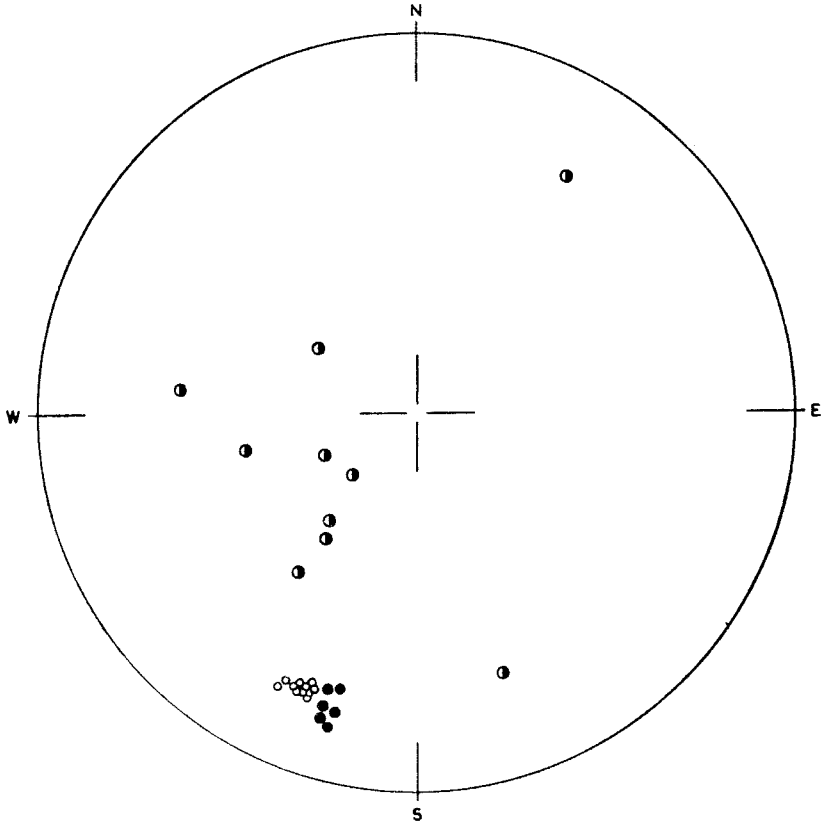


Polar (equal area) projection. Reversal at Shatterford.
× Upward dip.
● Downward dip.

Incidental to these researches was the discovery, at Shatterford, of igneous rocks, from two sites about 200 yards apart, whose directions of magnetization are plotted in fig. 3. They fall into two distinct groups, one having a mean polarization N 30° E with a downward dip of 3°,

and the other being magnetized N 215° E with an upward dip of 3°. The field evidence leaves no doubt that the rocks from both the sites belong to the same intrusion. The only possible explanation of this seems to be that the reversal must be due to some effect of the type postulated by Neel (1951). Heating tests performed so far in the laboratory give no indication that the mechanism of self-reversal has been preserved.

Fig. 4

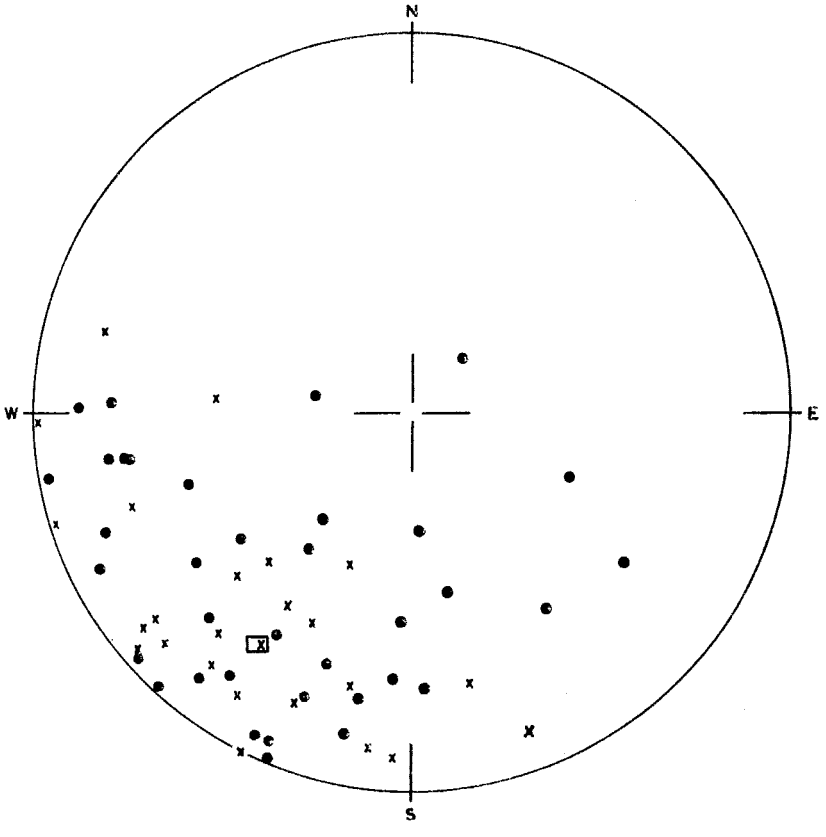


Polar (equal area) projection.
Country and Contact Rocks at Clee Hills.
◑ Unbaked country rock. Mean intensity $1{\cdot}6 \times 10^{-7}$ e.m.u./g.
● Underlying contact rock. Mean intensity 100×10^{-7} e.m.u./g.
○ Overlying contact rock. Mean intensity 300×10^{-7} e.m.u./g.
All dips downward.

A second point of interest is that it was possible, from our measurement, to resolve a long-standing controversy as to whether four particular igneous bodies, exposed at sites between the Clee Hills and Walsall, are intrusive or extrusive in origin. For while T. W. Pocock (1931) considered them to be lava flows, C. E. Marshall (1942) suggested that they may be sills. We collected samples of the igneous rock and of the overlying and

underlying sediments from each of these sites, and measured the directions of remanent magnetization. The results for one site (the Clee Hills) are shown in fig. 4. It is evident that while the unbaked sediments, from considerable distances away from the contact have scattered directions of magnetization, the rocks immediately above and below the igneous body are magnetized in similar directions, and have, moreover, a much higher intensity of magnetization. This seems to suggest strongly that they were both heated simultaneously, and therefore that the body is an intrusion. Similar evidence of intrusive origin has been found for the other three sites.

Fig. 5



Polar (equal area) projection. French Triassic sandstones.
× Upward dip.
● Downward dip.
□ Mean axis of British Trias (upwards).

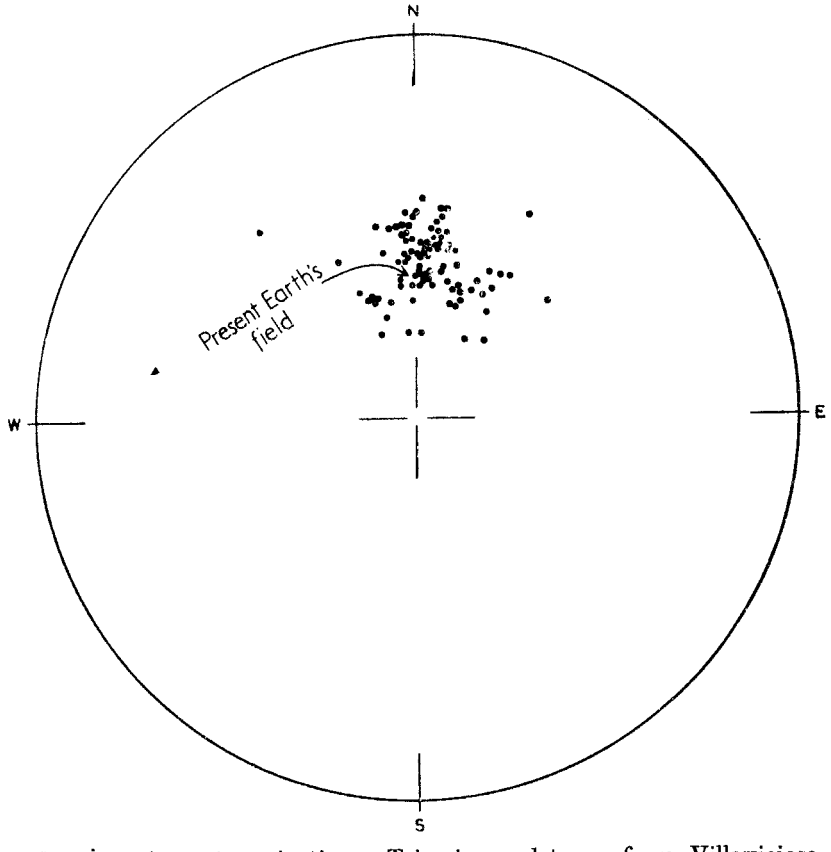
§ 2. Triassic Sandstones from France and Spain

It has been shown previously (Clegg, Almond and Stubbs 1954) that the rocks of the New Red Sandstone System in England are consistently

magnetized along a North East–South West axis, with shallow downward dips, suggesting that the British land mass has moved northward from close to the Equator, and rotated in a clockwise sense relative to the geographical poles, at some time since they were laid down. The measurements have now been extended to rocks of similar age on the continent of Europe.

Figure 5 shows the combined results of measurements on sandstones from seven sites in the Lower Trias of the Vosges region in France. It will be seen that the general North East–South West azimuthal direction of magnetization appears to persist on the Continent, although the scatter in direction is considerably greater than that observed in Britain, suggesting that the rocks may be only partially magnetically stable.

Fig. 6

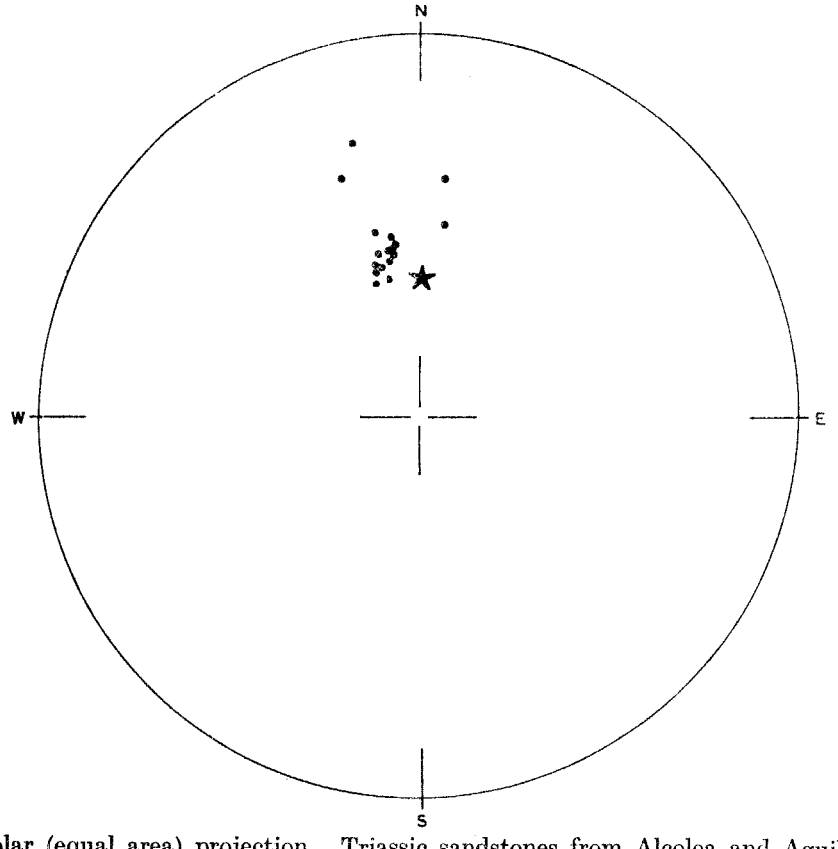


Polar (equal area) projection. Triassic sandstones from Villaviciosa.
(Uncorrected for geological tilt of strata.)
Downward dip.

In a paper published in 1954, S. W. Carey concluded from tectonic considerations that Spain may have undergone a rotation of some 35° in an anticlockwise sense during the Alpine orogeny, about a pivotal point situated at the Western end of the Pyrenees. It was considered

that a comparison of the magnetic directions of the Triassic sandstones of Spain and Great Britain might test this theory ; for if such a movement has occurred, Spanish rocks of the same age should be polarized along a North–South axis. Rocks were collected from a total of seven sites in North and Central Spain, but unfortunately those from four of these localities proved to be magnetically unstable. Of the three sites showing consistent magnetization that at Villiaviciosa, on the North coast near Oviedo, was most extensively sampled. The results are plotted in fig. 6, and it is evident that the mean direction lies close to that of the present Earth's field. The plots have not been corrected for the geological tilt

Fig. 7



Polar (equal area) projection. Triassic sandstones from Alcolea and Aguilar de Anguita. (Uncorrected for geological tilt of strata.)
• Downward dip.
★ Present Earth's field.

of the strata, which had a value of 30° and was directed within a few degrees of magnetic North, so that the magnetization of the samples when *in situ* would be almost vertically downwards. This suggests that the rocks possess some degree of magnetic stability. The magnetic

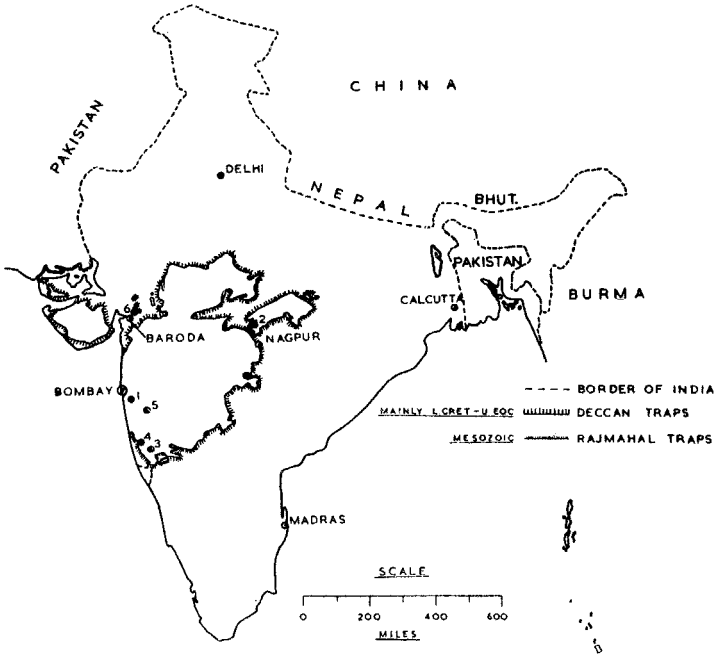
directions of rocks from two other sites, one near Alcolea in Central Spain and the other at Aguilar de Anguita nearby, are plotted together in fig. 7. They show similar polarizations, although the same test of stability cannot be applied to them since the strata were almost horizontal.

Comparison of these azimuthal directions of magnetization with those of the British Trias would appear to indicate that the postulated rotation of Spain has in fact occurred. However, since the mean magnetic inclination is approximately 25° steeper than that of the British rocks, while the mean latitude of the Spanish sites is some 10° farther South than that of the British, the final interpretation of the results remains inconclusive.

§ 3. Rocks from the Deccan Traps of India

Palaeomagnetic measurements on basaltic lavas from the Deccan Traps of Western India have been made previously by Irving (1956) who examined seven specimens from widely spaced localities, and by Clegg, Deutsch and Griffiths (1956) who obtained rather more than 400 specimens taken from two areas of limited lateral but considerable vertical extent, one in the neighbourhood of Khandala and the other near to Linga.

Fig. 8



Palaeomagnetic surveys (Deccan traps).
1, Khandala area. **2**, Linga area. **3**, Nipani area. **4**, Amba area. **5**, Kambatki Ghát. **6**, Pavagadh area.

Both sets of measurements showed the mean magnetic direction to lie along a NNW–SSE axis inclined rather steeply upwards towards the North, corresponding to a pole position close to Florida. The work of Clegg *et al.* has now been extended to include the following four extra localities (see fig. 8) in the Deccan Trap system :

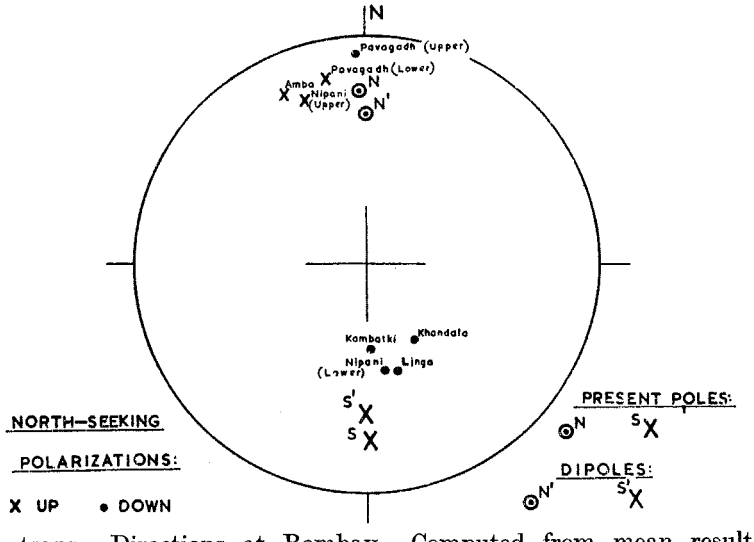
(*a*) the Nipani Area of Southern Bombay Province where specimens were collected from seven sites several miles apart ;

(*b*) the Amba area, close to Kolhapur City, where basalts were collected from the side of a ghat, over a range of elevations covering some 1500 feet ;

(*c*) the Kambatki area south of Poona ; and

(*d*) Mount Pavagadh, which rises abruptly 2400 feet above the surrounding plains to the north-east of Baroda in Gujerat. This hill is separated from the main body of the Deccan Traps by a distance of some 30 miles, and while on its lower slopes there are numerous exposures of horizontally bedded lavas typical of the Traps, these are overlain by 50 to 100 feet of acid tuffs, and capped by a massive plug or flow of rhyolite.

Fig. 9



Deccan traps. Directions at Bombay. Computed from mean results for various areas.

The mean directions of magnetization of the rocks from the various areas are shown in fig. 9 and are listed, together with other relevant data, in table 2. The results for the lower mainly basic flows and the upper acid rocks of the Pavagadh area have been treated separately, and the same procedure has been adopted for the lower and upper flows of the Nipani and Amba areas where the higher have opposite polarizations from the lower. The magnetic directions for the original Khandala

and Linga areas (Clegg, Deutsch and Griffiths 1956) are also shown in the figure.

Table 2

(1) Area	(2) No. of Speci-mens	(3) Intensity of magn. $\times 10^3$ c.g.s.u./c.c.	(4) Mean azimuthal direction α	(5) Mean Dip δ	(6) Radius of 50% circle of confidence
Kambatki	5	2·7	176°	+60° down	2·5°
Nipani (lower)	44	1·9	170°	+57° down	4·5°
Nipani (upper)	59	1·6	340°	−30° up	3·6°
Amba (lower)	54	3·1	141°	+59° down	4·1°
Amba (upper)	34	2·9	335°	−23° up	4·5°
Pavagadh (lower)	41	3·2	348°	−12° up	4·1°
Pavagadh (upper acid)	17	0·3	357°	+15° down	3·5°

It is evident that the present results are in broad agreement with those obtained previously by Clegg *et al.*, and that they are compatible with the assumption that the Indian sub-continent has undergone a substantial northward movement relative to the Earth's geographical axis at some time since late Cretaceous or early Eocene times when the rocks were formed. There is, however, a considerable difference between the mean magnetic directions found at the various sites. This is clearly illustrated in fig. 10, where the ancient pole positions corresponding to the values of α and δ given in table 2 are plotted, together with that given by Irving (1956).

It is interesting to note that the pole positions for the lower flows of the Nipani, Amba and Pavagadh areas lie farther from the present geographical pole than do those for the upper rocks. These discrepancies may evidently arise from one of a number of causes, viz :

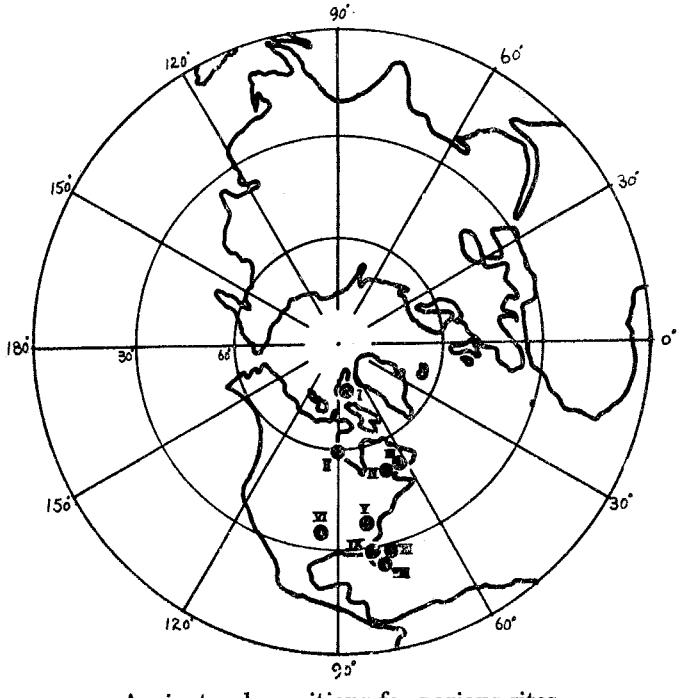
(*a*) They may be due to some physical phenomenon associated with different depths of burial, such, for example, as that described by Graham (1957).

(*b*) They may be indicative of partial magnetic instability, and the consequent pulling of the magnetic directions towards that of the present Earth's field. For rocks which are magnetized normally (N.N.W. with upward dips) this would tend to reduce the magnetic dip, and consequently to move the apparent position of the ancient pole northwards, while for those magnetized in the reverse direction it would have the opposite effect. It is worth noting that the four most northerly pole positions (Nos. 1, 2, 3 and 4 in fig. 10) were obtained from normally magnetized rocks, and that the remainder which lie at considerable distances from the present pole, are from rocks with reverse magnetization.

(*c*) They may represent a northward movement of India relative to the Earth's rotational axis, during the time when the Deccan Traps were being laid down.

This work is being further extended and a more detailed account and discussion of the results will appear shortly elsewhere.

Fig. 10



Ancient pole positions for various sites.
I, Pavagadh (upper). II, Pavagadh (lower). III, Amba (upper). IV, Nipani (upper). V, Linga. VI, Nipani (lower). VII, Amba (lower). VIII, Khandala. IX, Irving's (1956) pole position from seven sites.

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